

Top confusion pairs (test set, 59 total errors)

True	Predicted	Count	Why?
4	9	5	Open loop vs. stem
5	3	5	Similar curvature
9	4	4	Open loop vs. stem
7	2	4	Diagonal bar confusion
×	7	3	Cross $\approx$ slanted stroke

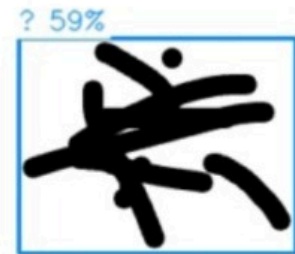
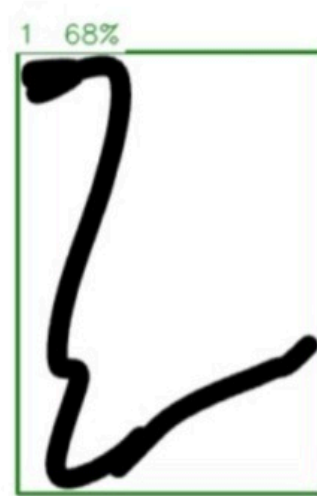
Key observations

- × is the weakest class: 94.8% accuracy (6 errors in 116)
- – and ÷: 100% accuracy on the test set
- Most errors are **digit-to-digit**, not cross-type
- Worst pairs are visually near-identical shapes

With only 1 048 operator samples, × is most sensitive to style variation — being confused as a slanted 7.

Solver needs single characters

- probability vector gets converted to char
- confidence > 60% → decision
- otherwise return “?”



Solver

- chars merged to string expression
- expression solved with AST evaluator

1. Draw $3 + 4 \rightarrow$ verify result 7
2. Draw $12 \times 3 \rightarrow$ verify 36
3. Draw $8 \div 2 \rightarrow$ verify 4
4. Draw an ambiguous symbol $\rightarrow$ show ? chip
5. Point out per-symbol color coding

```
streamlit run draw.py
```

```
Model: outputs/best_model.keras · 14 classes
```

App screenshot

99.44% Test accuracy

14 Unified classes

178K Parameters

< 3 min Training time

What we built — End-to-end math solver running in the browser.

- One unified CNN — digits **and** operators
- Safe AST evaluator (no `exec` / `eval`)
- OpenCV segmentation with proximity merging

Adding a new symbol class = more training data only, no new code.

Future work — Next steps to extend the project.

- Multi-digit numbers as a single unit
- Fractions and parentheses
- More classes: =, decimal point, variables
- Larger operator dataset → reduce failure cases
- Mobile / touch input optimization

Thank You

Questions?

Matti H. V. Lehmann

Philip Nguyen

David A. Spickenheuer

Staniya Thomas

Jonas Schenke

Abhirup Sain